

ULTRON 2.7 Phase 4 Execution Report

Optional LLM planner integration

Project	ULTRON 2.7
Phase	Phase 4: Optional LLM Planner
Workspace	Local workspace checkout: U2.7
Status	Implemented and verified

Executive Summary

Phase 4 adds an optional LLM planner integration while preserving ULTRON's tool-based safety model. The model can propose a JSON tool call, but schema validation, risk policy, confirmation gates, and executor restrictions remain authoritative.

Phase 4 Scope

- Add a reusable LLM planner interface.
- Add an Ollama provider using the local /api/chat endpoint.
- Parse and validate JSON tool-call output.
- Add CLI and config controls for rules, hybrid, and LLM planner modes.
- Fail closed when the provider is unavailable or the model returns invalid output.

What Was Implemented

Area	File(s)	Result
LLM adapter	src/ultron27/llm.py	Prompt builder, JSON parser, LLMPlanner, and Ollama provider.
Configuration	src/ultron27/config.py	planner_mode, llm_provider, llm_model, llm_endpoint, llm_timeout_seconds.
CLI	src/ultron27/cli.py	--planner-mode, --llm-model, --llm-endpoint, runtime LLM trace.
Docs	README.md, docs/architecture.md, docs/phase4_llm_planner.md	Explains the optional LLM planner and safety gates.
Tests	tests/test_pipeline.py	Expanded to 21 regression tests.

Planner Modes

Mode	Behavior	Use
rules	Dataset, regex, and fuzzy planning only.	Default mode; works offline.
hybrid	Use rules first, then try LLM when no safe rule matches.	Best first LLM mode.
llm	Try LLM first; fall back safely on provider or validation failure.	LLM-focused experiments.

Verification Results

Check	Command	Result
Unit tests	<code>python -m unittest discover -s tests</code>	21 tests passed.
Dataset evaluation	<code>python scripts/evaluate_dataset.py --split test</code>	Intent 1.0, tool 1.0, argument exact match 0.9614, confirmation 1.0.
Safety regression	<code>python scripts/evaluate_safety_regression.py</code>	362 cases, tool accuracy 1.0, policy action accuracy 1.0.
LLM unavailable behavior	<code>python -m ultron27 ... --planner-mode llm</code>	Provider error recorded; command fell back and was blocked.

Recommended Next Step

- Run Ollama locally with the configured model and test hybrid mode against ambiguous commands.
- Add an LLM fixture evaluation set with expected tool calls.
- Measure JSON validity, tool accuracy, argument accuracy, and clarification quality.
- Only after stable LLM planning, move to Phase 5 voice input.